

Vahe Gdlyan

ML Engineer & Data Scientist | Production Systems | Computer Vision | Statistical Experimentation

Yerevan, Armenia · gdlyanvahe31@gmail.com · linkedin.com/in/vahe-gdlyan · github.com/VaheGdlyan · medium.com/@gdlyanvahe31

SUMMARY

ML engineer and data scientist with a deployed YOLOv8 precision-agriculture system, a custom loss architecture, and a mathematical OOD analysis paper. Built and validated production-grade statistical experimentation pipelines and a two-stage revenue prediction system across 887K and 390K user datasets respectively. Co-built Vestimate — a multi-cloud distributed AI wardrobe platform spanning FastAPI, Redis/Celery, Modal serverless GPU inference, Supabase pgvector, and a Flutter cross-platform client — from architecture to deployed alpha in one month. Technical writer publishing ML and engineering frameworks on Medium.

EDUCATION & TRAINING

Picsart Academy — Machine Learning & Data Science Trainee (Production ML Curriculum)

Yerevan, Armenia | Feb. 2025 – Present

Abovyan Educational Complex — High School; Focus on Mathematics and Computer Science

Abovyan, Armenia | Expected Graduation: 2026

- Extracurricular: 2023 Vice Champion of Armenia in Junior Boxing.

ENGINEERING PROJECTS

Vestimate: Multi-Cloud AI Wardrobe Intelligence Platform | FastAPI, Flutter, Redis, Celery, Modal.com, Supabase, FashionCLIP, U-2-Net

Lead Architect | Jan. 2025 – Present

- Architected a full distributed ML system end-to-end: FastAPI async gateway (202 Accepted pattern), Redis/Celery asynchronous task queue, Modal.com serverless NVIDIA T4 GPU inference, Supabase PostgreSQL with pgvector extension, and Cloudflare R2 object storage — operational at ≤\$80/month via scale-to-zero infrastructure design.
- Designed a dual-model GPU inference pipeline containerising U-2-Net (nested encoder-decoder salient object detection) and FashionCLIP (ViT-B/16 contrastive vision-language transformer, 2.5M+ fashion pairs) in a single Modal container; warm inference latency ~1.2s per garment.
- Engineered a 512-dimensional semantic wardrobe vector store using pgvector IVFFlat cosine indexing (100 lists); cosine similarity queries execute in <15ms across wardrobes of 1,000+ items.
- Built a dual-stage RAG recommendation engine: Stage 1 pure pgvector cosine retrieval; Stage 2 gpt-4o-mini contextual synthesis constrained to pre-validated candidate sets with a Python hallucination guard cross-validating every UUID before client delivery — 95% semantic consistency rate.
- Implemented a zero-shot FashionCLIP classification system using cosine distance over a compiled natural language anchor dictionary (category, material, fit, 32 chromatic anchors); 85% pass rate at 0.70 confidence threshold with automatic ML Confidence Gate routing to manual_review_queue.
- Built Fernet symmetric encryption for Google Calendar OAuth tokens at rest; integrated Open-Meteo real-time weather API; constructed composite context query vectors weighting weather and calendar signals for situational outfit synthesis.
- Designed a Redis cache layer (Index 1) achieving <80ms response on repeat daily opens (98%+ hit rate), bypassing LLM inference entirely on warm requests.
- Implemented graceful degradation fallback: pure pgvector color-harmony + category matching when third-party services (OpenAI, Google) are unreachable — Flutter client never crashes.
- Built Flutter cross-platform client using Clean Architecture, Riverpod 3.0 reactive state, Hive offline persistence layer, and native camera integration for garment digitisation.

AgriVision: End-to-End CV Decision Support System | YOLOv8, PyTorch

Jan. 2026 – Present | [GitHub]

- Designed focal boundary loss with adaptive photometric weighting; deliberately traded marginal recall to eliminate 1,092 zero-IoU hallucinated detections, preventing a 4–8% yield overestimation.
- Maintained 0.944 mAP@50 against the Global Wheat holdout partition while increasing median prediction confidence by 39.4% and reducing required training epochs by 30%.
- Authored a 7-page mathematical diagnostic paper detailing Out-of-Distribution boundary analysis (preparing for arXiv preprint submission).
- Deployed the final inference state engine via a Hugging Face cloud server to compute spatial uniformity.

DATA SCIENCE & ANALYTICS PROJECTS

A/B Test Stress-Test Pipeline | Python, SciPy, Pandas

Mar. 2025 – Apr. 2025 | [GitHub] [Medium]

- Engineered a 5-gate statistical validation pipeline across 887K users, preventing the deployment of a false-positive conversion lift (+4.29%, $p=0.1669$) by requiring significance survival across independent tests.
- Detected and named the "False Flat" failure mode: a +4.29% relative lift with a 95% CI crossing zero, indistinguishable from noise under a two-proportion Z-test; flagged as a hollow win before shipping.
- Applied Welch's T-test over Student's T-test for Revenue per Visitor analysis on a zero-inflated distribution (98%+ of users spent \$0), surfacing a statistically significant +8.17% RPV lift hidden by the primary metric.
- Conducted Simpson's Paradox pressure-testing via weekday/weekend segmentation (63.5%/36.5% split); confirmed structural stability with a 0.21pp lift delta across behaviorally distinct environments.
- Applied Gini coefficient analysis to quantify revenue concentration risk; Variant B reduced inequality from 0.307 to 0.291, proving the lift was distributed across the user base rather than driven by whale spenders.

Zero-Inflated Revenue Prediction: Two-Stage Hurdle Model | LightGBM, XGBoost, Scikit-learn

[GitHub] [Medium]

- Architected a two-stage Hurdle Model on 390K users (1.69% payer rate) decomposing revenue prediction into $P(\text{Payer}) \times E[\text{Revenue} | \text{Payer}]$; achieved 92.9% Revenue Capture Rate in the top predicted decile.
- Built Stage 1 LightGBM classifier with `scale_pos_weight=58.1`, GOSS sampling, and stratified 5-fold CV; established logistic regression baseline (ROC-AUC 0.943, AP 0.245) as a scientific control.
- Diagnosed and corrected systematic 4x probability over-confidence via Platt Scaling calibration, reducing MAE from \$8.79 to \$1.83 (79.2% improvement) with zero degradation to ranking performance.
- Engineered 107 behavioral features from a 72-hour user lifecycle window; identified saves per session and paid interest segment as primary payer-class separators through permutation importance analysis.
- Applied log-transform regression ($\log_{1p}$) for Stage 2 XGBoost Valuator on a multi-modal revenue distribution (\$14/\$53/\$79 tiers); designed threshold optimisation across 86 candidate values to minimise production MAE.

Product Funnel & User Behavior Analysis | Python, Pandas, Matplotlib, Seaborn

Feb. 2025 – Mar. 2025 | [GitHub]

- Performed end-to-end funnel analysis quantifying conversion drop-off across acquisition, activation, and monetisation stages to identify highest-leverage product intervention points.
- Segmented user cohorts by behavioral and demographic attributes to surface heterogeneous conversion patterns masked by aggregate funnel metrics.
- Produced actionable drop-off diagnostics translating statistical findings into prioritised product recommendations, establishing the analytical foundation for the subsequent A/B testing pipeline.

TECHNICAL WRITING & KNOWLEDGE SHARING

UnderTheHood Engineering | Medium Publication

medium.com/@gdlyanvahe31 | Apr. 2025 – Present

- Authoring production-grade ML and experimentation frameworks as flagship technical articles, bridging academic statistical methodology and engineering decision-making for a professional audience.
- "The A/B Test Stress-Test Pipeline" — Documents the full 5-gate experimental validation framework including SRM auditing, zero-inflation handling, Simpson's Paradox testing, and Gini coefficient risk analysis.
- "Your Model Has a 92.9% Score and It's Still Lying to You" — Details the calibration failure mode in imbalanced ML systems and the production fix via Platt Scaling, with full mathematical derivations.

AWARDS & RECOGNITION

■ 1st Place — Plekhanov Russian University of Economics, Yerevan Branch Startup & Innovation Competition | May 2025

Yerevan, Armenia | Competed against university teams presenting concepts, mockups, and research prototypes.

- Presented Vestimate as a live, operational multi-cloud AI system — the only entry in the competition running production infrastructure at the time of judging. Competing teams presented concepts, UI mockups, and theoretical proposals; Vestimate demonstrated a fully containerised distributed pipeline spanning six cloud platforms with measurable latency metrics and a working Flutter client.
- Judged across technical depth, commercial viability, and execution maturity by a panel of academics, industry practitioners, and investors. Vestimate's combination of deployed infrastructure, quantified performance characteristics (<80ms cache hits, <4s cold-path end-to-end, 95% LLM semantic consistency), and a credible B2B commercialisation roadmap produced a unanimous first-place result.
- Recognised specifically for the execution gap between Vestimate and all competing entries: the only team that shipped an architecture rather than described one.

TECHNICAL SKILLS

Languages: Python, SQL, Dart

ML Frameworks & Tools: PyTorch, LightGBM, XGBoost, Scikit-learn, Hugging Face Transformers, FashionCLIP, U-2-Net, YOLOv8, Git, Linux, Modal.com

Backend & Infrastructure: FastAPI, Redis, Celery, Supabase (PostgreSQL + pgvector), Cloudflare R2, Railway, Docker, REST API design, asynchronous task queue architecture, scale-to-zero serverless GPU infrastructure

Frontend & Mobile: Flutter (Clean Architecture, Riverpod 3.0, Hive offline persistence), Dart

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, SciPy

Analytics & Experimentation: A/B testing, statistical significance testing (Z-test, Welch's T-test, Chi-square), Sample Ratio Mismatch (SRM) detection, Simpson's Paradox analysis, Gini coefficient concentration analysis, funnel analysis, cohort segmentation, causal inference, metric decomposition, false positive prevention frameworks

ML Architectures: CNNs, RNNs, LSTMs, YOLO, Gradient Boosting, Logistic Regression, Vision Transformers (ViT), Contrastive Learning (CLIP), Dual-stage RAG, Retrieval-Augmented Generation, pgvector IVFFlat cosine indexing

Core Competencies: Custom loss function design, model calibration (Platt Scaling), revenue modelling, zero-inflated distribution handling, out-of-distribution analysis, distributed systems architecture, production AI infrastructure, vector database design, hallucination-resistant LLM pipeline engineering